Comparative Analysis of Supervised Learning Algorithms for Spam Detection

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I. INTRODUCTION

This experiment evaluates the computational and predictive performance of three supervised learning algorithms for spam identification on the Spambase dataset. This study analyzes algorithms based on training time, accuracy, and F1-scores. The analysis includes divided ten-fold cross-validation and Friedman and Nemenyi statistical tests to evaluate differences in performance.

II. DATASET

The Spambase dataset, from the UCI Machine Learning Data Repository, has 4601 instances and 58 attributes. The features include word and character frequencies, as well as capital letter data. The target variable determines whether an email is spam (1 or 0).

III. ALGORITHMS

Several supervised learning techniques were investigated, including:

1.Logistic Regression (LR): It is a linear model for binary classification.
2.Random Forest (RF): Random Forest is an ensemble learning method using decision trees.
3.Support Vector Machines (SVM): A kernel-based algorithm for classification.

IV. EXPERIMENTAL SETUP

1.Preprocessing:

- Checked for missing values and none found.
- Using StandardScaler, features were scaled for Logistic Regression and SVM.

2.Cross-validation:

- We made divided ten-fold cross-validation to guarantee balanced class distribution across folds.

3.Performance metrics:

- Training Time: A measure of computing efficiency.
- Accuracy: Accuracy indicates the total correctness of a prediction.
- F1-Score: Balances precision and recall to address class imbalance.

4.Statistical tests:

- Friedman test: The Friedman test identifies statistically significant rank imbalances.
- Nemenyi Test: The Nemenyi Test identifies algorithms with significant variations.

V. RESULTS

Accuracy Table

TABLE I
PERFORMANCE OF LR, RF, AND SVM ACROSS FOLDS.

Fold	LR	RF	SVM
Fold 1	0.9154	0.9523	0.9197
Fold 2	0.9217	0.9565	0.9196
Fold 3	0.9283	0.9609	0.9326
Fold 4	0.9326	0.9674	0.9370
Fold 5	0.9087	0.9457	0.9174
Fold 6	0.9370	0.9587	0.9413
Fold 7	0.9348	0.9565	0.9304
Fold 8	0.9261	0.9565	0.9283
Fold 9	0.9391	0.9435	0.9391
Fold 10	0.9109	0.9522	0.9065
Avg	**0.9255**	**0.9550**	**0.9272**
Std	**0.0104**	**0.0067**	**0.0106**

This table shows the accuracy of three supervised learning algorithms Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) across ten folds in a cross-validation setting. The table shows the accuracy numbers for each fold, as well as the average (Avg) accuracy and standard deviation (Std) of each algorithm. Based on data:

- Random Forest (RF) has the highest average accuracy (0.9550) and the smallest standard deviation (0.0067), showing excellent predictive performance and consistency.

Fold	Logistic Regression	Random Forest	SVM
Fold 1	0.9154 (3)	0.9523 (1)	0.9197 (2)
Fold 2	0.9217 (2)	0.9565 (1)	0.9196 (3)
Fold 3	0.9283 (3)	0.9609 (1)	0.9326 (2)
Fold 4	0.9326 (3)	0.9647 (1)	0.9370 (2)
Fold 5	0.9087 (3)	0.9457 (1)	0.9174 (2)
Fold 6	0.9370 (3)	0.9587 (1)	0.9413 (2)
Fold 7	0.9348 (2)	0.9565 (1)	0.9043 (3)
Fold 8	0.9261 (3)	0.9522 (1)	0.9283 (2)
Fold 9	0.9391 (2)	0.9455 (1)	0.9391 (2)
Fold 10	0.9109 (2)	0.9522 (1)	0.9065 (3)
Avg Rank	2.6	1.0	2.4

TABLE II
COMPARISON OF LOGISTIC REGRESSION, RANDOM FOREST, AND SVM

- SVM and LR perform well, although with somewhat lower accuracy and more variability than RF.

F1 Score Table

TABLE III
F1-SCORE COMPARISON OF LR, RF, AND SVM MODELS ACROSS 10 FOLDS.

Fold	LR	RF	SVM
Fold 1	0.8914	0.9389	0.8975
Fold 2	0.8989	0.9441	0.8964
Fold 3	0.9076	0.9500	0.9136
Fold 4	0.9136	0.9580	0.9197
Fold 5	0.8846	0.9311	0.8956
Fold 6	0.9197	0.9471	0.9256
Fold 7	0.9153	0.9448	0.9096
Fold 8	0.9029	0.9438	0.9065
Fold 9	0.9186	0.9266	0.9191
Fold 10	0.8839	0.9375	0.8796
Avg	0.9036	0.9422	0.9063
Std	0.0129	0.0086	0.0134

The table shows the F1-scores of Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) using tenfold cross-validation. RF has the highest average F1-score (0.9422) and the lowest variability (Std: 0.0086), indicating outstanding and consistent performance. In comparison, SVM (0.9063) and LR (0.9036) have similar scores but larger variability. Overall, RF has higher accuracy than the other models.

Fold	Logistic Regression	Random Forest	SVM
Fold 1	0.8914 (3)	0.9389 (1)	0.8975 (2)
Fold 2	0.8989 (2)	0.9441 (1)	0.8963 (3)
Fold 3	0.9076 (3)	0.9500 (1)	0.9136 (2)
Fold 4	0.9136 (3)	0.9580 (1)	0.9197 (2)
Fold 5	0.8846 (3)	0.9311 (1)	0.8956 (2)
Fold 6	0.9197 (3)	0.9471 (1)	0.9197 (2)
Fold 7	0.9153 (2)	0.9448 (1)	0.9095 (3)
Fold 8	0.9023 (3)	0.9438 (1)	0.9126 (2)
Fold 9	0.9186 (3)	0.9266 (1)	0.9191 (2)
Fold 10	0.8839 (2)	0.9375 (1)	0.8796 (3)
Avg Rank	2.7	1.0	2.3

TABLE IV
F1 SCORE RESULTS WITH RANKS

Training Time Table

TABLE V
TRAINING TIME COMPARISON (SECONDS)

Fold	LR	RF	SVM
Fold 1	0.0124	0.3664	0.2182
Fold 2	0.0080	0.3630	0.2297
Fold 3	0.0076	0.3576	0.1985
Fold 4	0.0062	0.3536	0.2761
Fold 5	0.0085	0.3516	0.2172
Fold 6	0.0082	0.3585	0.2288
Fold 7	0.0071	0.3438	0.2325
Fold 8	0.0066	0.3586	0.2880
Fold 9	0.0077	0.3570	0.2193
Fold 10	0.0064	0.3499	0.2185
Avg	**0.0079**	**0.3560**	**0.2327**
Std	**0.0017**	**0.0062**	**0.0264**

The table defines the training time of Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) using tenfold cross-validation. LR has the shortest training time (Avg: 0.0079s) and lowest deviation (Std: 0.0017), making it the fastest. RF has much longer training times (Avg: 0.3560s), but with little variability (SD: 0.0062). Training times for SVM are moderate (average: 0.2327s), while variability is important (standard deviation: 0.0264).

Fold	Logistic Regression	Random Forest	SVM
Fold 1	0.0150 (1)	0.6041 (3)	0.3774 (2)
Fold 2	0.0150 (1)	0.6080 (3)	0.3950 (2)
Fold 3	0.0104 (1)	0.6116 (3)	0.3443 (2)
Fold 4	0.0095 (1)	0.6061 (3)	0.4035 (2)
Fold 5	0.0096 (1)	0.6027 (3)	0.3730 (2)
Fold 6	0.0103 (1)	0.6047 (3)	0.3933 (2)
Fold 7	0.0107 (1)	0.6078 (3)	0.4012 (2)
Fold 8	0.0103 (1)	0.5956 (3)	0.4011 (2)
Fold 9	0.0108 (1)	0.6078 (3)	0.3686 (2)
Fold 10	0.0092 (1)	0.5989 (3)	0.3775 (2)
Avg Rank	1.0	3.0	2.0

TABLE VI
TRAINING TIME RESULTS WITH RANKS

Statistical Analysis

- Friedman Test: Used to determine whether the average ranks of algorithms differ significantly.

$$Q = \frac{12n}{k(k+1)} \sum_{j=1}^{k} \left(\bar{r}_j - \frac{k+1}{2} \right)^2$$

In this equation, n is the number of folds, k is the number of algorithms, and R is the average rank (R j).

- Nemenyi Test: Used when the Friedman Test reveals significant differences. It calculates the crucial difference (CD):

$$CD = q\alpha \sqrt{\frac{k(k+1)}{6n}}$$

If the total difference between ranks exceeds the CD, the algorithms differ significantly.

VI. CONCLUSION

Random Forest exceeds Logistic Regression and SVM in terms of accuracy and F1 Score, but required much more computation time.
Logistic Regression was the quickest approach, making it ideal for time-sensitive applications.
Statistical testing demonstrated significant differences, with Random Forest outperforming Logistic Regression in predictive performance and computing efficiency.

Predictive performance (accuracy and F1 score):

- Algorithm 1 exceeds Algorithm 2.
- Algorithm 3 performs similarly to Algorithm 1 and outperforms Algorithm 2.

Training Time:

- Algorithm 1 requires much more training time compared to Algorithm 2.
- Algorithms 2 and 3 have excellent processing efficiency. Algorithm 1 is the most accurate but most computationally expensive.

Practical Implications:

- Algorithm 3 strikes a suitable compromise between accuracy, F1 score, and training duration.
- Algorithm 2 is computationally efficient but fails to perform in both predictive metrics.

VII. REFERENCES

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